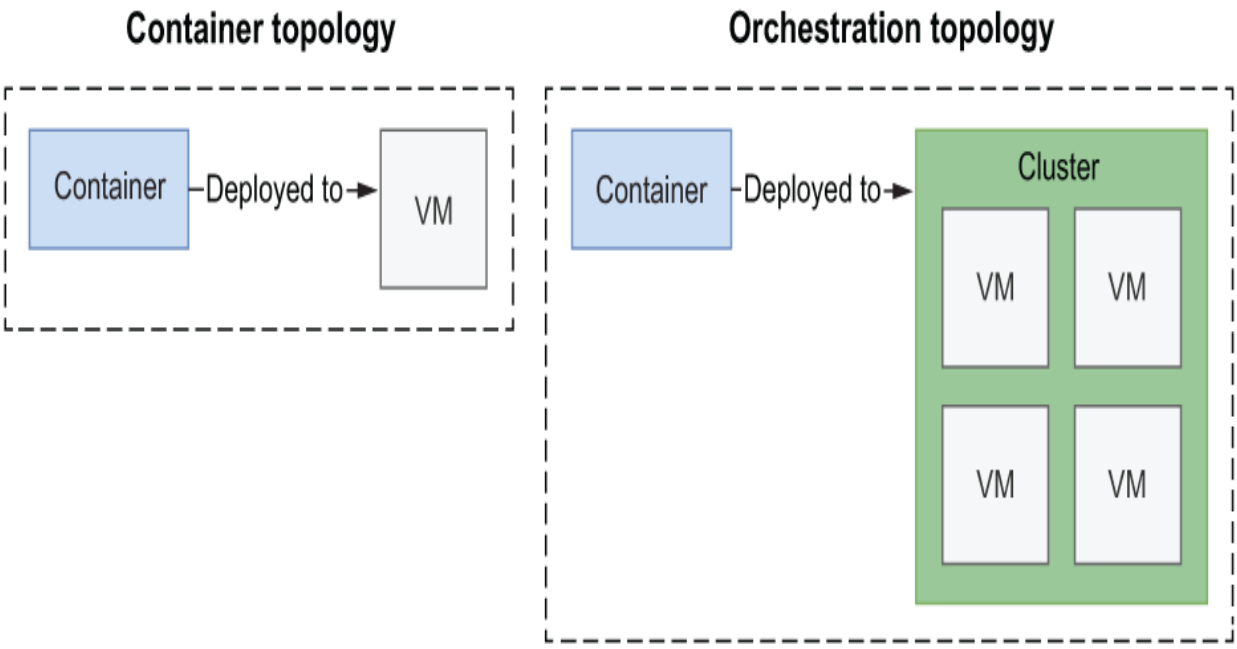


Figure 1.11 The deployment target of containers is a machine, whereas for orchestrators, it's a cluster.

Container orchestration helps you automate many different tasks:

- Managing clusters, bringing up and down machines when necessary
- Scheduling and deploying containers within a cluster to a machine that meets the container requirements for CPU and memory
- Dynamically scaling containers for high availability and resilience, leveraging health monitoring
- Setting up networks for containers to communicate with each other, defining routing, service discovery, and load balancing
- Exposing services to the internet, establishing ports and networks
- Allocating resources to containers according to specific criteria
- Configuring the applications running within the containers